Final Lab Report: ITN 7.02 – Case Study Part 4 (Modules 11 – 13)

Step 1 – Designing and documenting an IPv4 subnet scheme

Using the provided IPv4 address range 192.168.1.0 /24, the subnet scheme is calculated using VLSM:

LAN	Required Hosts	Allocated Hosts	Network Address	CIDR / Mask	Range	Broadcast
LAN-A	60	62	192.168.1.0	/26 (255.255.255.192)	192.168.1.1–62	192.168.1.63
LAN-B	25	30	192.168.1.64	/27 (255.255.255.224)	192.168.1.65–94	192.168.1.95
LAN-C	20	30	192.168.1.96	/27 (255.255.255.224)	192.168.1.97–126	192.168.1.127
LAN-D	12	14	192.168.1.128	/28 (255.255.255.240)	192.168.1.129–142	192.168.1.143
LAN-E	12	14	192.168.1.144	/28 (255.255.255.240)	192.168.1.145–158	192.168.1.159
Link 1	2	2	192.168.1.160	/30 (255.255.255.252)	192.168.1.161–162	192.168.1.163
Link 2	2	2	192.168.1.164	/30 (255.255.255.252)	192.168.1.165–166	192.168.1.167
Link 3	2	2	192.168.1.168	/30 (255.255.255.252)	192.168.1.169–170	192.168.1.171
Link 4	2	2	192.168.1.172	/30 (255.255.255.252)	192.168.1.173–174	192.168.1.175

Step 2, 5 & 6 – Router Configuration (NET_1_R1)

```
None
!
hostname NET_1_R1
!
ipv6 unicast-routing
!
interface GigabitEthernet0/0/0
 description Connection to LAN-1 (PCs)
 ipv6 address fe80::1 link-local
```

```
 ipv6 address 2001:ABCD:1234:1::1/64
 no shutdown
!
interface GigabitEthernet0/0/1
 description Connection to LAN-2 (Admin)
 ipv6 address fe80::1 link-local
 ipv6 address 2001:ABCD:1234:2::1/64
 no shutdown
!
end
```

Step 3 & 6 – PC IPv6 Addressing Table

Host	GUA	Link-Local	Default Gateway
PC1	2001:ABCD:1234:1::15/64	FE80::15	2001:ABCD:1234:1::1
PC2	2001:ABCD:1234:1::16/64	FE80::16	2001:ABCD:1234:1::1
PC3	2001:ABCD:1234:1::17/64	FE80::17	2001:ABCD:1234:1::1
PC4	2001:ABCD:1234:1::18/64	FE80::18	2001:ABCD:1234:1::1
PC5	2001:ABCD:1234:1::19/64	FE80::19	2001:ABCD:1234:1::1
Admin1	2001:ABCD:1234:2::15/64	FE80::15	2001:ABCD:1234:2::1
Admin2	2001:ABCD:1234:2::16/64	FE80::16	2001:ABCD:1234:2::1

Step 7 – Presentation Theory Questions

1. **Subnet Mask Role:** Defines which part of the 32-bit IPv4 address is the Network ID vs Host ID.
2. **Determining Network:** Hosts use binary AND operations between their IP and subnet mask to determine their network boundary.
3. **Transmission Types:**
    - Unicast: One-to-one communication.
    - Multicast: One-to-many communication (group).
    - Broadcast: One-to-all communication on a local network segment.
4. **Why IPv6:** Exhaustion of IPv4 address space and the need for a larger address space for IoT and global connectivity.

5. **GUA vs Link-Local:** GUA is globally routable on the internet. Link-local is for local segment communication (FE80::/10) and is not routable.
6. **GUA Structure:** Global Routing Prefix (Organization identification), Subnet ID (internal network division), Interface ID (specific host identifier).
7. **Ping & Tracert:** Ping tests host-to-host connectivity using ICMP Echo. Tracert is used to trace the path packets take through routers to troubleshoot packet loss or delays at specific hops.